

Evaluation of Cyber security and the Internet of Things: Vulnerabilities, threats, intruders, and attacks by Abomhara and Køien (2015)

Alan B. Palayil
University of the Cumberland
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Dr. Michael Vubinski
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Article Evaluation:

Abomhara, M., & Køien, G. M. (2015). *Cyber security and the Internet of Things: Vulnerabilities, threats, intruders, and attacks*. *Journal of Cyber Security and Mobility*, 4(1), 65–88.

Introduction

The Internet of Things (IoT) has changed the way people, businesses, and governments use technology. Billions of connected devices, from health wearables to industrial systems, promise to make things more efficient and newer. But this growth also brings new and difficult cybersecurity problems. Conventional perimeter-based models fail to sufficiently safeguard decentralized, resource-limited IoT ecosystems.

To tackle this situation, Abomhara and Køien (2015) investigated the IoT's unique vulnerabilities, threats, and attack vectors. Their paper provides a classification of risks, identifies adversaries, and suggests countermeasures tailored to the constraints of the Internet of Things (IoT). This evaluation looks at their research question, investigation, conclusions, and overall contribution. It also suggests other points of view and future directions.

Authors' Investigation and Approach

Abomhara and Køien (2015) performed a narrative literature review instead of an empirical investigation. They used information from academic papers, standards, and industry reports to make a **taxonomy of vulnerabilities, intruders, and attacks**.

The article talked about **IoT's unique features** that make risks worse: its diversity, mobility, limited devices, and huge scale. For instance, sensors with few resources often cannot manage strong encryption, and mobile nodes are easy to spoof and listen in on. They put **threat actors** into four groups: curious amateurs, malicious insiders, cybercriminals, and nation-states. They said that each group has different goals and skills.

Finally, the authors talked about **countermeasures** like lightweight cryptography, secure identity management, monitoring, and intrusion detection. All of these are related to privacy, authentication, confidentiality, integrity, and availability (CIA). Their suggestions were meant to link theory to practice for real-world IoT security.

Research Question

Although the article did not explicitly state a formal hypothesis, its guiding research question can be inferred:

What are the key vulnerabilities, threat actors, and attack vectors in IoT environments, and what security requirements and countermeasures are necessary to mitigate them?

The implied hypothesis is that IoT requires **tailored, end-to-end security frameworks**, as traditional IT defenses cannot address IoT's dynamic and distributed nature.

Summary of the Article

The article offers a **comprehensive taxonomy** of IoT security issues, structured as follows:

1. **IoT Characteristics:** Scale, heterogeneity, limited resources, and ad-hoc connectivity.
 2. **Vulnerabilities:** Weak authentication, insecure firmware, poor isolation, and unencrypted communications.
 3. **Threat Actors:** Ranging from hobbyists to nation-states.
 4. **Attacks:**
 - Device level: tampering, malware.
 - Network level: eavesdropping, spoofing, denial of service.
 - Application level: data misuse, privacy violations.
 5. **Security Requirements:**
 - Confidentiality: encryption and access control.
 - Integrity: hashing, signatures, secure boot.
 - Availability: redundancy, DoS detection.
 - Privacy: anonymization, storage protection.
 - Authentication: identity-based and multi-factor methods.
 6. **Recommendations:** Lifecycle-based security, continuous monitoring, international standards, and collaboration between academia and industry.
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Authors' Conclusions

The authors concluded that IoT **increases the digital attack surface**, which makes perimeter-based defenses useless. IoT needs **protection that covers the whole lifecycle**, from design to retirement instead.

They said that **privacy is especially important** for user trust because the IoT collects personal and behavioral data. Standardization and collaboration were also important because broken solutions make global ecosystems weak.

In the end, they said that IoT needs custom solutions like lightweight cryptography, scalable key management, monitoring, and secure updates to stay safe from new threats.

Do the Data Support the Conclusions?

The article functions as a narrative review, drawing on existing literature instead of empirical validation. The conclusions are **conceptually substantiated** and correspond with extensive research. Roman et al. (2013) also talked about the problems that IoT has when it is spread out, and Sicari et al. (2015) focused on privacy-by-design. These affirm that Abomhara and Køien's assertions align with scholarly consensus.

However, the absence of a systematic methodology constrains rigor. No inclusion criteria, search strategy, or quantitative synthesis are specified, which creates the potential for selection bias. While sufficient for agenda-setting, more robust evidence necessitates meta-analyses, breach datasets, or experimental IoT testbeds.

Nonetheless, taxonomy offers significant conceptual clarity and continues to be instrumental in guiding further investigation.

Alternative Explanations

Several nuances complicate the authors' conclusions:

1. **Heterogeneity as resilience:** IoT diversity may reduce monoculture risks, preventing single vulnerabilities from scaling globally.
2. **Sectoral differences:** Risks differ across industries; industrial IoT often has stricter controls than consumer IoT.
3. **Economic pressures:** Many vulnerabilities stem from cost and usability trade-offs, not purely technical gaps. Regulations and incentives may be as important as technology.

4. **Human factors:** User behavior, such as poor password hygiene or neglected updates, amplifies risks and deserves greater emphasis.
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Other Approaches and Future Research

Future studies could build upon Abomhara and Køien's taxonomy in several ways:

1. **Systematic Reviews:** Transparent review protocols to assess prevalence of threats and defenses.
 2. **Empirical Data:** Longitudinal studies of IoT breaches across industries.
 3. **Testbeds:** Experimental environments to compare lightweight PKI, DTLS/OSCORE, and blockchain-based identity.
 4. **Privacy Research:** Field evaluations of techniques like differential privacy or federated learning.
 5. **Convergence Studies:** Integrating physical and cyber monitoring for IoT security (Fausto et al., 2021).
 6. **Economic Analysis:** Cost–benefit studies to encourage adoption of robust protections.
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Relevance and Importance of the Study

The article is still relevant. IoT is now the backbone of healthcare, smart cities, and industrial systems, where failures can have serious safety and financial effects. Even though it came out in 2015, people still use taxonomy as a **basic framework**. Subsequent studies, including Sicari et al. (2015), have corroborated its fundamental conclusions.

Even though technology has gotten better with 5G, AI, and edge computing, the vulnerabilities that were found are still important. The study's lasting significance resides in delineating the problem space and directing researchers and practitioners toward holistic solutions.

Appropriateness of the Design

For 2015, a narrative review was appropriate to scope a developing field. It effectively highlighted critical issues and set a research agenda.

By current standards, however, the design has limits. Narrative reviews are prone to bias and lack transparency. A systematic review would have offered stronger evidence, and empirical data would help validate recommendations.

Still, as an exploratory, agenda-setting work, the design was sufficient to contribute meaningfully to the early IoT security literature.

Conclusion

Abomhara and Køien (2015) created an important list of IoT vulnerabilities, threats, and intruders. Their work made the risks clearer and showed how important it is to have security that works throughout the whole life cycle. Their conclusions are consistent with subsequent scholarship and continue to hold relevance, although their dependence on narrative review diminishes empirical robustness.

Different points of view, like how diversity can make people stronger or how economic pressures can make people feel insecure, add depth to their claims. Future research should concentrate on systematic reviews, empirical datasets, and experimental testbeds to validate and enhance their framework.

The article is an important piece of work overall. It does not see IoT security as a problem that has been solved; instead, it sees it as an ongoing challenge that needs innovative ideas in technology, organization, and regulation.

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